pH and indicators

20a

Indicators

Acids and bases change the colour of certain dyes. These dyes are called indicators. Indicators tell us whether a substance is acidic, basic or neutral. Some indicators in the school laboratory are kept as liquids in dropper bottles, while others are dried onto strips of paper. Litmus, phenolphthalein, methyl orange and bromothymol are some common indicators. Indicators can also be made from plants, for example, the liquid from boiled rhubarb stems turns red in acids and green in bases.

pH scale

Scientists have an accurate way of measuring acidity. They use a very sensitive indicator called pH paper. The pH paper changes into one of several colours when dipped into a liquid substance. The colour is matched to numbers on a pH scale. The pH scale gives very acidic solutions the number 1, very basic solutions number 14, and neutral solutions number 7.

Diagram 1: Some substances and their pH numbers

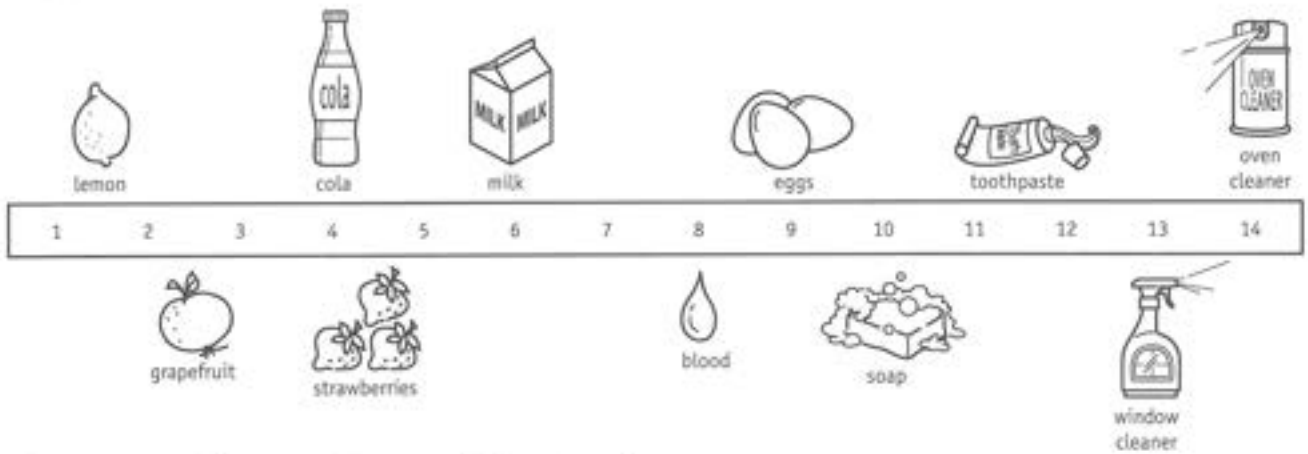


Diagram 2: Indicator colours and the pH scale

pH	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	strongly acidic			weakly acidic			neutral		weakly basic				strongly	basic
litmus	|			red						blue				|
methyl orange	| red								yellow					|
pH paper	|	red		|	yellow		pale yellow			green		blue	|
phenolphthalein	|				colourless			pink				red-purple		|
thymol blue	|	red		|		yellow			green		blue		|
universal	| red			orange			yellow	green		blue		purple	deep purple	

Table 1: Plant indicators and colours

Plant source	Colour of indicator solution		
	as extracted	*in acids*	*in bases*
hydrangea	colourless	pink	yellow
pansy	purple	pink	yellow
pink geranium	pink	pink	orange
red cabbage	purple	red	green
rhubarb	red	red	pale green
rose (pink)	yellow	pink	orange
sweet pea	light green	pink	yellow

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pH and indicators

20b

Use the text, diagrams and table above to answer the following questions.

1 State the colour of universal indicator for the common substances in diagram 1. Record your answers in the table below.

Substance	Colour of universal indicator

2 State the colour of litmus if the plant indicators in Table 1 are either orange, red or yellow. Tabulate your results.

3 Five unknown solutions were tested with indicators by a group of students. Their results are shown below.

Solution	Indicator	Colour
A	pH paper	pale yellow
B	methyl orange	red
C	universal	purple
D	thymol blue	green
E	litmus	blue

a Name the acidic solutions. _____

b List the basic solutions. _____

c State the neutral solutions. _____

4 The students were then to test each solution A, B, C, D and E with the plant indicators red cabbage, rhubarb and rose. Show what their results should be in the table below.

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pH and indicators

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5 The teacher provided the group of students with an unknown solution F. The students' task was to find the pH of the solution. They tested the solution with several indicators as shown in their results table.

Indicator	Colour
hydrangea	pink
methyl orange	yellow
phenolphthalein	colourless
thymol blue	yellow

Deduce the pH range of solution F. _____

6 a List the solutions A, B, C, D and F in order of their acidity.

Most acidic _____ Most basic

b Explain why the solution E was not included in the question above.

_____

7 a Explain why labels on cleaning substances for ovens and drains warn the user to wear gloves and not to swallow the substances.

_____

_____

b Describe what you would do if some cleaning substance splashed into your eye.

_____

8 A gardener measures some garden soil and finds that it has a pH of 5.2. Since the flowers planted there grow best at a pH of 7, the gardener adds some crushed limestone to the soil. Explain whether limestone is acidic, neutral or basic.

_____

_____

_____

9 Using your knowledge of the pH scale complete the table below, listing any foods, drinks and household products you use regularly in the correction columns.

Acidic	Neutral	Basic

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